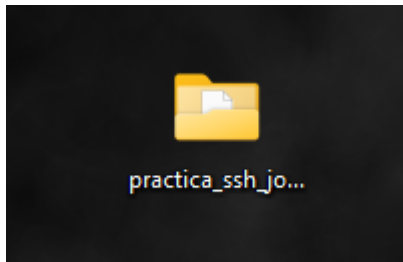


Linux V - Bastionado ssh

1. Despliegue de Infraestructura (Vagrant)

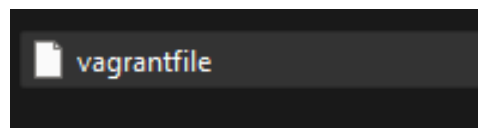
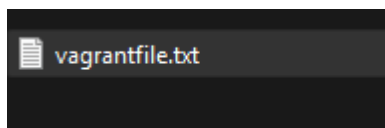
Primero vamos a crear una carpeta en el escritorio (por ejemplo) llamada Pratica_ssh



Dentro de esta carpeta vamos a hacer un txt llamado vagrantfile y vamos a pegar esto

```
Vagrant.configure("2") do |config|
  config.vm.box = "ubuntu/jammy64"
  config.vm.network "private_network", ip: "192.168.56.10"
  config.vm.hostname = "srv-seguro"
  config.vm.provider "virtualbox" do |vb|
    vb.memory = "2048"
    vb.cpus = 2
  end
end
```

Una vez creado vamos a eliminar la extensión txt borramos todo menos el nombre y el .



Una vez todo esto listo vamos a abrir un CMD y nos vamos a ir a la carpeta donde tengamos el archivo creado y pondremos el siguiente comando: **vagrant up**

```
C:\Users\calav\Desktop\practica_ssh_jorge>vagrant up
Bringing machine 'default' up with 'virtualbox' provider...
==> default: Importing base box 'ubuntu/jammy64'...
==> default: Matching MAC address for NAT networking...
==> default: Checking if box 'ubuntu/jammy64' version '20241002.0.0' i
==> default: Setting the name of the VM: practica_ssh_jorge_default_17
==> default: Clearing any previously set network interfaces...
==> default: Preparing network interfaces based on configuration...
    default: Adapter 1: nat
    default: Adapter 2: hostonly
==> default: Forwarding ports...
    default: 22 (guest) => 2222 (host) (adapter 1)
==> default: Running 'pre-boot' VM customizations...
==> default: Booting VM...
==> default: Waiting for machine to boot. This may take a few minutes.
    default: SSH address: 127.0.0.1:2222
    default: SSH username: vagrant
    default: SSH auth method: private key
    default:
```

Una vez finalice el proceso escribimos el comando: **vagrant ssh**
Y entraremos a la maquina virtual ubuntu con el usuario vagrant

```
vagrant@srv-seguro: /home  X + v
C:\Users\calav\Desktop\practica_ssh_jorge>vagrant ssh
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-164-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Jan  8 15:59:50 UTC 2026
```

```
vagrant@srv-seguro:~$ ls
vagrant@srv-seguro:~$ pwd
/home/vagrant
vagrant@srv-seguro:~$ cd home
-bash: cd: home: No such file or directory
vagrant@srv-seguro:~$ cd /home/
vagrant@srv-seguro:/home$ ls
ubuntu  vagrant
vagrant@srv-seguro:/home$
```

2. Hardening del sServicio SSH

2.1. Gestión de Claves (Host -> Guest)

Desde nuestra máquina vamos a generar claves SSH para ello vamos a usar este comando: `ssh-keygen -t rsa -b 4096 -f key_bastionado`

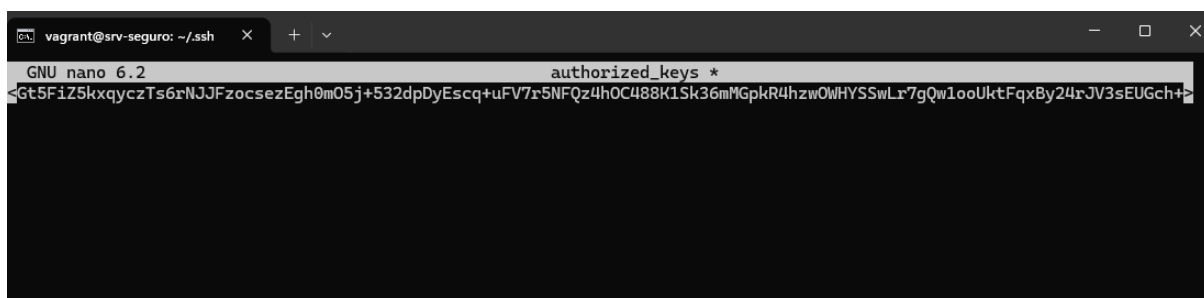
```
C:\Users\calav>ssh-keygen -t rsa -b 4096 -f key_bastionado
Generating public/private rsa key pair.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in key_bastionado
Your public key has been saved in key_bastionado.pub
The key fingerprint is:
SHA256:v2/ny4Cu5k4ALs8Ld35GXUZhdBzsNu/v58SY9C6Qqbc calav@DESKTOP-VFRJDQ0
The key's randomart image is:
+---[RSA 4096]-----+
|          .++o. |
|          ...o  |
|          .    |
|          .    |
|          o +   |
|          S. o +o|
|          +   .+. =|
|          . + ... o...o.+|
|          o + .+...o0000|
|          . .*+.o+E++=B|
+---[SHA256]-----+

C:\Users\calav>
```

Ahora en la máquina vagrant vamos a copiar la claves para ello vamos a la siguiente ruta y hacemos un `sudo nano` a `authorized_keys`

```
vagrant@srv-seguro:/$ cd /home/vagrant/.ssh/
vagrant@srv-seguro:~/.ssh$ ls
authorized_keys
vagrant@srv-seguro:~/.ssh$ sudo nano authorized_keys
```

Aquí pegaremos nuestra clave generada anteriormente (para poder ver la clave hacemos un `type key_bastionado.pub` y hay veremos la clave) debemos reemplazarla por la que habrá dentro del archivo

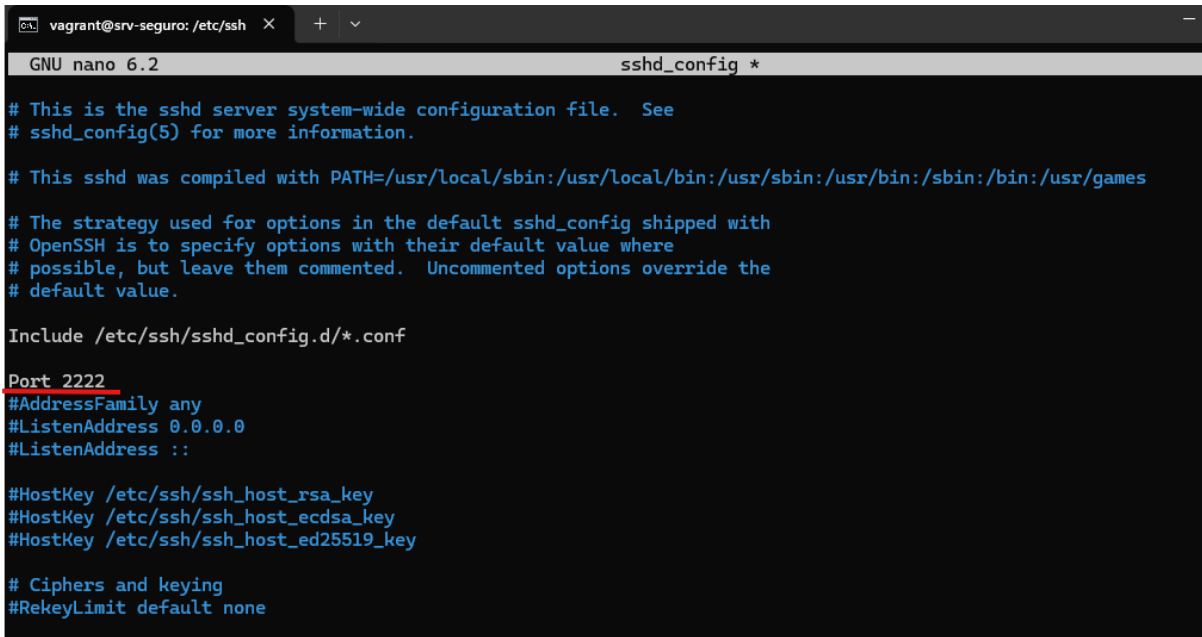


```
vagrant@srv-seguro: ~/.ssh
GNU nano 6.2                                authorized_keys *
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQBAQC...Gt5FiZ5kxqyczTs6rNJJFzocsezEgh0m05j+532dpDyEscq+uFV7r5NFQz4h0C488K1Sk36mMGpkR4hzw0WlHYSSmLr7gQw1ooUktFqxBy24rJV3sEUGch+
```

2.2. Configuración de sshd_config.

Ahora vamos a cambiar algunas directivas para el bastionado de SSH. Para ello vamos a editar el siguiente archivo `/etc/ssh/sshd_config`.

Como primer cambio vamos a mover el puerto de ssh 22 al 2222.



```
vagrant@srv-seguro: /etc/ssh x + v
GNU nano 6.2 sshd_config *
# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

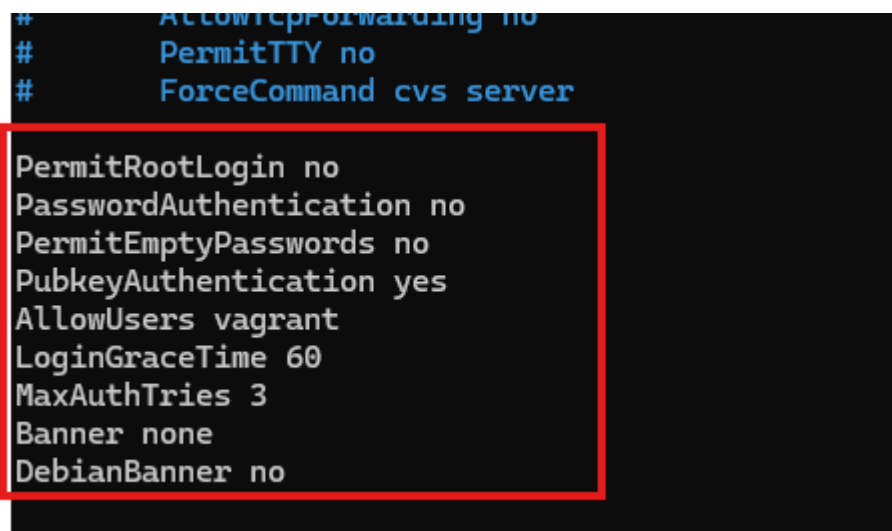
Port 2222
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none
```

También añadimos los siguientes parámetros al final del archivo:

PermitRootLogin no	# o Prohibir root.
PubkeyAuthentication yes	# o Permitir llaves.
PasswordAuthentication no	# o Prohibir contraseñas.
PermitEmptyPasswords no	# o Prohibir contraseñas vacías.
AllowUsers vagrant	# o Lista blanca estricta.
LoginGraceTime 60	# o Tiempo máximo para loguearse de 60s.
MaxAuthTries 3	# o Evitar fuerza bruta.
Banner none	# o Ocultar banner personalizado.
DebianBanner no	# o Ocultar versión del OS en el banner.



```
# AllowTcpForwarding no
# PermitTTY no
# ForceCommand cvs server

PermitRootLogin no
PasswordAuthentication no
PermitEmptyPasswords no
PubkeyAuthentication yes
AllowUsers vagrant
LoginGraceTime 60
MaxAuthTries 3
Banner none
DebianBanner no
```

2.3. Aplicación y Verificación

Para poder aplicar los siguientes cambios y comprobarlos primero vamos a reiniciar ssh y comprobar su estado

Como comprobamos el servicio está activo y está escuchando en el puerto 2222

```
vagrant@srv-seguro:/etc/ssh$ sudo systemctl restart sshd
vagrant@srv-seguro:/etc/ssh$ sudo systemctl status sshd
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Thu 2026-01-08 16:45:43 UTC; 5s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
  Process: 2276 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
 Main PID: 2277 (sshd)
    Tasks: 1 (limit: 2307)
   Memory: 1.7M
      CPU: 19ms
  CGroup: /system.slice/ssh.service
          └─2277 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jan 08 16:45:43 srv-seguro systemd[1]: Starting OpenBSD Secure Shell server...
Jan 08 16:45:43 srv-seguro sshd[2277]: Server listening on 0.0.0.0 port 2222.
Jan 08 16:45:43 srv-seguro sshd[2277]: Server listening on :: port 2222.
Jan 08 16:45:43 srv-seguro systemd[1]: Started OpenBSD Secure Shell server.
vagrant@srv-seguro:/etc/ssh$
```

Para comprobar que podemos entrar por ssh desde nuestra máquina vamos a abrir otro CMD y vamos a colocar el siguiente comando

`ssh -i key_bastionado -p 2222 vagrant@192.168.56.10`

```
C:\Users\calav>ssh -i key_bastionado -p 2222 vagrant@192.168.56.10
The authenticity of host '[192.168.56.10]:2222 ([192.168.56.10]:2222)' can't be established.
ED25519 key fingerprint is SHA256:RGcdSwZ8uzRjAxZEBxV4efiCe4gMsTZnWqndzJ3jHA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[192.168.56.10]:2222' (ED25519) to the list of known hosts.
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-164-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Jan  8 16:49:33 UTC 2026

System load:          0.0
Usage of /:            4.0% of 38.70GB
Memory usage:         11%
Swap usage:           0%
Processes:            104
Users logged in:      1
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:79:64ff:feba:195

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.3 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jan  8 16:00:18 2026 from 10.0.2.2
vagrant@srv-seguro:~$
```

3. Despliegue del Honeypot

3.1. Preparación de Dependencias

Vamos a instalar las dependencias necesarias para realizar un

`sudo apt update / upgrade`

```
vagrant@srv-seguro:/etc/ssh$ sudo apt update
vagrant@srv-seguro:/etc/ssh$ sudo apt upgrade
```

3.2. Instalación de Cowrie

Antes de instalar el honeypot vamos a crear el usuario cowrie

```
vagrant@srv-seguro:/etc/ssh$ sudo adduser --disabled-password cowrie
Adding user 'cowrie' ...
Adding new group 'cowrie' (1002) ...
Adding new user 'cowrie' (1002) with group 'cowrie' ...
Creating home directory '/home/cowrie' ...
Copying files from '/etc/skel' ...
Changing the user information for cowrie
Enter the new value, or press ENTER for the default
  Full Name []: cowrie
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n] y
vagrant@srv-seguro:/etc/ssh$
```

Cuando lo creamos vamos a cambiar a ese usuario

```
vagrant@srv-seguro:/etc/ssh$ sudo su - cowrie
cowrie@srv-seguro:~$
```

Clonamos el repositorio de github

```
cowrie@srv-seguro:~$ git clone http://github.com/cowrie/cowrie
Cloning into 'cowrie'...
warning: redirecting to https://github.com/cowrie/cowrie/
remote: Enumerating objects: 20690, done.
remote: Counting objects: 100% (1013/1013), done.
remote: Compressing objects: 100% (494/494), done.
remote: Total 20690 (delta 887), reused 519 (delta 519), pack-reused 19677 (from 4)
Receiving objects: 100% (20690/20690), 11.39 MiB | 13.26 MiB/s, done.
Resolving deltas: 100% (14348/14348), done.
cowrie@srv-seguro:~$
```

Cuando lo clonemos vamos a entrar a la carpeta del cowrie y vamos a activar / crear el entorno de python virtualizado y también actualizamos los paquetes de python

```
cowrie@srv-seguro:~$ cd cowrie/
cowrie@srv-seguro:~/cowrie$ virtualenv --python=python3 cowrie-env
created virtual environment CPython3.10.12.final.0-64 in 437ms
  creator CPython3Posix(dest=/home/cowrie/cowrie/cowrie-env, clear=False, no_vcs_ignore=False, global=False)
  seeder FromAppData(download=False, pip=bundle, setuptools=bundle, wheel=bundle, via=copy, app_data_dir=/home/cowrie/.local/share/virtualenv)
  added seed packages: pip==22.0.2, setuptools==59.6.0, wheel==0.37.1
  activators BashActivator, CShellActivator, FishActivator, NushellActivator, PowerShellActivator, PythonActivator
cowrie@srv-seguro:~/cowrie$ source cowrie-env/bin/activate
(cowrie-env) cowrie@srv-seguro:~/cowrie$ pip install --upgrade pip
Requirement already satisfied: pip in ./cowrie-env/lib/python3.10/site-packages (22.0.2)
Collecting pip
  Downloading pip-25.3-py3-none-any.whl (1.8 MB)
    1.8/1.8 MB 8.3 MB/s eta 0:00:00
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 22.0.2
    Uninstalling pip-22.0.2:
      Successfully uninstalled pip-22.0.2
Successfully installed pip-25.3
(cowrie-env) cowrie@srv-seguro:~/cowrie$
```

```
(cowrie-env) cowrie@srv-seguro:~/cowrie$ pip install --upgrade -r requirements.txt
Collecting attrs==25.4.0 (from -r requirements.txt (line 1))
```

3.3. Configuración del Señuelo

Ahora vamos a editar el siguiente archivo para poder cambiar el nombre del host por alguno que parezca atractivo para un atacante y el puerto de escucha del cowrie

```
cowrie@srv-seguro: ~/cowrie X + v
GNU nano 6.2 cowrie.cfg *
# =====
# General Cowrie Options
# =====
[honeypot]

# Sensor name is used to identify this Cowrie instance. Used by the database
# logging modules such as mysql.
#
# If not specified, the logging modules will instead use the IP address of the
# server as the sensor name.
#
# (default: not specified)
#sensor_name=myhostname

# Hostname for the honeypot. Displayed by the shell prompt of the virtual
# environment
#
# (default: svr04)
hostname = srv-finanzas

# Directory where to save log files in.
#
# (default: log)
log_path = var/log/cowrie

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute   ^C Location   M-U Undo      M
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify   ^_/ Go To Line  M-E Redo      M
```

```
[ssh]

# Enable SSH support
# (default: true)
enabled = true
listen_port = 2223
```

Ahora iniciaremos cowrie primero vamos a colocar este comando `pip install -e .`

```
(cowrie-env) cowrie@srv-seguro:~/cowrie$ pip install -e .
Obtaining file:///home/cowrie/cowrie
Installing build dependencies ... done
Checking if build backend supports build_editable ... done
Getting requirements to build editable ... done
Installing backend dependencies ... done
Preparing editable metadata (pyproject.toml) ... done
```


Y ya podremos iniciar cowrie con el comando **cowrie start**

```
(cowrie-env) cowrie@srv-seguro:~/cowrie$ cowrie start
Join the Cowrie community at: https://www.cowrie.org/slack/

Starting cowrie: [twistd --umask=0022 --pidfile /home/cowrie/cowrie/var/run/cowrie.pid --logger cowrie.python.logfile.lo
gger cowrie]...
/home/cowrie/cowrie/cowrie-env/lib/python3.10/site-packages/twisted/conch/ssh/transport.py:110: CryptographyDeprecationW
arning: TripleDES has been moved to cryptography.hazmat.decrepit.ciphers.algorithms.TripleDES and will be removed from c
ryptography.hazmat.primitives.ciphers.algorithms in 48.0.0.
  b"3des-cbc": (algorithms.TripleDES, 24, modes.CBC),
/home/cowrie/cowrie/cowrie-env/lib/python3.10/site-packages/twisted/conch/ssh/transport.py:117: CryptographyDeprecationW
arning: TripleDES has been moved to cryptography.hazmat.decrepit.ciphers.algorithms.TripleDES and will be removed from c
ryptography.hazmat.primitives.ciphers.algorithms in 48.0.0.
  b"3des-ctr": (algorithms.TripleDES, 24, modes.CTR),
(cowrie-env) cowrie@srv-seguro:~/cowrie$
```

3.4. Redirección de puertos

Vamos a aplicar una regla con iptables que para quien intente conectar con el puerto 22 caiga automáticamente al puerto 2223 que es donde tenemos el honeypot

Para ello vamos a poner el comando **exit** para salir del usuario del cowrie

```
(cowrie-env) cowrie@srv-seguro:~/cowrie$ exit
Logout
vagrant@srv-seguro:/etc/ssh$ cd ..
```

Una vez fuera vamos a aplicar la siguiente regla

sudo iptables -t nat -A PREROUTING -p tcp --dport 22 -j REDIRECT --to-port 2223

Una vez aplicada comprobamos que se a creado correctamente

```
vagrant@srv-seguro:/$ sudo iptables -t nat -A PREROUTING -p tcp --dport 22 -j REDIRECT --to-port 2223
vagrant@srv-seguro:/$ sudo iptables -t nat -L -n -v
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source    destination
    0      0 REDIRECT  tcp  --  *      *       0.0.0.0/0  0.0.0.0/0          tcp dpt:22 redir ports 2223

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source    destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source    destination

Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source    destination
vagrant@srv-seguro:/$
```

4. Auditoría y Análisis

4.1. Prueba de Ataque (Desde el Anfitrión)

Desde mi maquina vamos a entrar al puerto real y veremos como podemos entrar perfectamente usaremos este comando

```
ssh -i key_bastionado -p 2222 vagrant@192.168.56.10
```

```
C:\Users\calav>ssh -i key_bastionado -p 2222 vagrant@192.168.56.10
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-164-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Jan  8 18:01:04 UTC 2026

System load:          0.01
Usage of /:           5.5% of 38.70GB
Memory usage:        17%
Swap usage:          0%
Processes:           111
Users logged in:      1
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:79:64ff:feba:195

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

New release '24.04.3 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jan  8 17:41:09 2026 from 192.168.56.1
vagrant@srv-seguro:~$
```

Ahora probaremos la conexión por ssh normal (puerto 22)

Como podemos ver nos pedirá una contraseña colocamos la que sea y nos dejara entrar como podemos ver esto seria el honeypot

```
C:\Users\calav>ssh root@192.168.56.10
root@192.168.56.10's password:

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@srv-finanzas:~# ls
root@srv-finanzas:~#
```

4.2. Análisis Forense

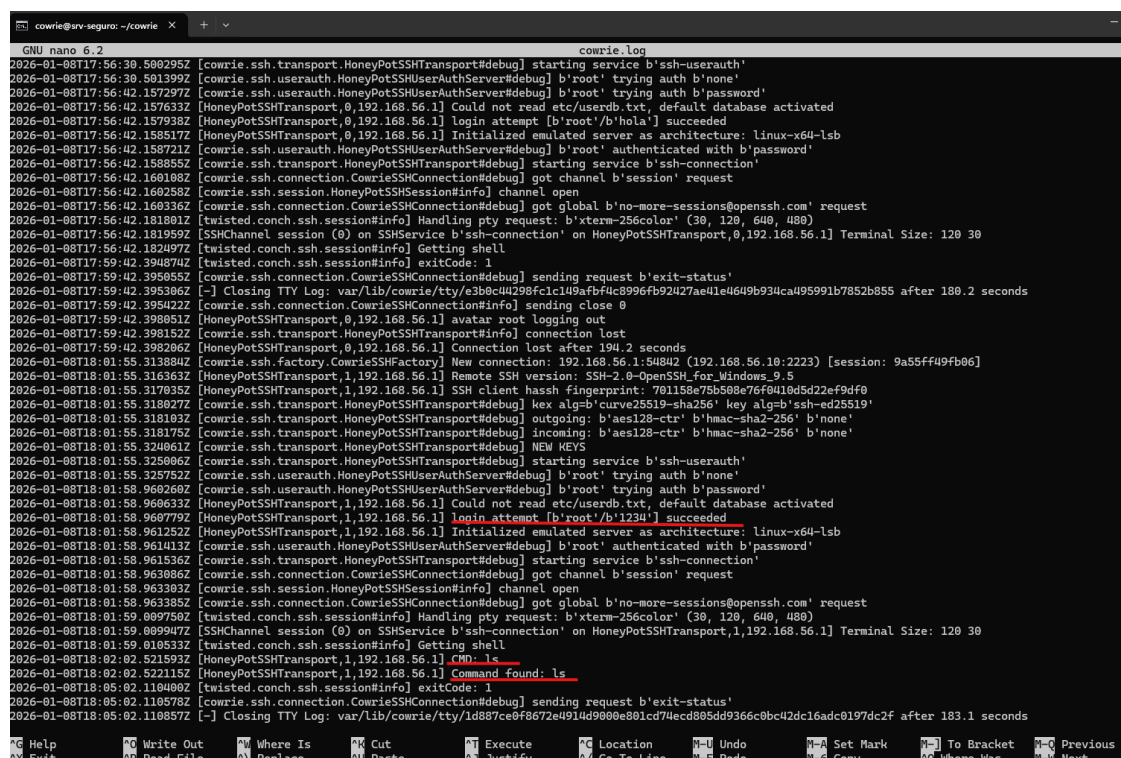
Dentro del usuario de cowrie vamos a la siguiente ruta

[cowrie/var/log/cowrie](#)

Dentro de aquí tendremos 2 archivos que serán los logs aquí podremos ver lo intentos de acceso al honeypot / usuarios / contraseña que se usó y los comandos que el atacante usará

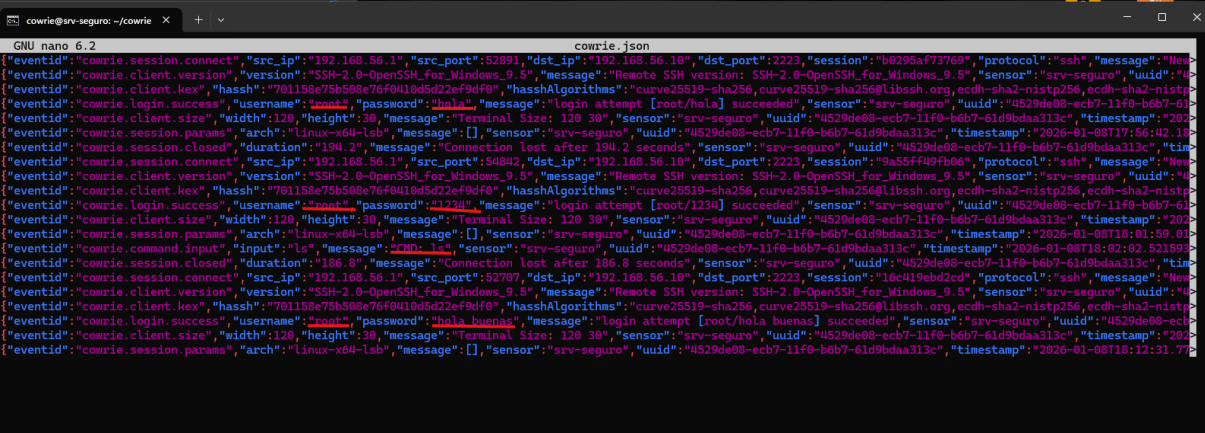
```
cowrie@srv-seguro:~/cowrie/var/log$ cd cowrie/
cowrie@srv-seguro:~/cowrie/var/log/cowrie$ ls
cowrie.json  cowrie.log
```

Si hacemos una nano a cowrie.log podremos ver este archivo que aparecerá todo lo mencionado anteriormente



```
GNU nano 6.2 cowrie.log
2026-01-08T17:56:30.500295Z [cowrie.ssh.transport.HoneyPotSSHTransport#debug] starting service b'ssh-userauth'
2026-01-08T17:56:30.501399Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'root' trying auth b'none'
2026-01-08T17:56:42.157297Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'root' trying auth b'password'
2026-01-08T17:56:42.157633Z [HoneyPotSSHTransport,0,192.168.56.1] Could not read etc/userdb.txt, default database activated
2026-01-08T17:56:42.157938Z [HoneyPotSSHTransport,0,192.168.56.1] Login attempt [b'root'/b'hola'] succeeded
2026-01-08T17:56:42.158517Z [HoneyPotSSHTransport,0,192.168.56.1] Initialized emulated server as architecture: linux-x64-lsb
2026-01-08T17:56:42.158721Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'root' authenticated with b'password'
2026-01-08T17:56:42.158855Z [cowrie.ssh.transport.HoneyPotSSHTransport#debug] starting service b'ssh-connection'
2026-01-08T17:56:42.160108Z [cowrie.ssh.connection.CowrieSSHConnection#info] got channel b'session' request
2026-01-08T17:56:42.160258Z [cowrie.ssh.session.HoneyPotSSHSession#info] channel open
2026-01-08T17:56:42.160336Z [cowrie.ssh.connection.CowrieSSHConnection#debug] got global b'no-more-sessions@openssh.com' request
2026-01-08T17:56:42.181801Z [twisted.conch.ssh.session#info] Handling pty request: b'xterm-256color' (30, 120, 640, 480)
2026-01-08T17:56:42.305065Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,0,192.168.56.1] Terminal Size: 120 30
2026-01-08T17:56:42.182497Z [twisted.conch.ssh.session#info] Getting shell
2026-01-08T17:56:42.304970Z [twisted.conch.ssh.session#info] exitCode: 1
2026-01-08T17:56:42.305065Z [cowrie.ssh.connection.CowrieSSHConnection#debug] sending request b'exit-status'
2026-01-08T17:56:42.395306Z [-] Closing TTY Log: var/lib/cowrie/tty/43b8cd4298fclcl48afbf4c8996fb92427ae41e4649b934ca495991b7852b855 after 180.2 seconds
2026-01-08T17:56:42.395422Z [cowrie.ssh.connection.CowrieSSHConnection#info] sending close 0
2026-01-08T17:56:42.398051Z [HoneyPotSSHTransport,0,192.168.56.1] avatar root logging out
2026-01-08T17:56:42.398152Z [cowrie.ssh.transport.HoneyPotSSHTransport#info] connection lost
2026-01-08T17:56:42.398266Z [HoneyPotSSHTransport,0,192.168.56.1] Connection lost after 194.2 seconds
2026-01-08T18:01:55.313844Z [cowrie.ssh.factory.CowrieSSHFactory] New connection: 192.168.56.1:54842 (192.168.56.10:2223) [session: 9a55fff49fb06]
2026-01-08T18:01:55.316363Z [HoneyPotSSHTransport,1,192.168.56.1] Remote SSH version: SSH-2.0-OpenSSH_for_Windows_9.5
2026-01-08T18:01:55.317035Z [HoneyPotSSHTransport,1,192.168.56.1] SSH client hash fingerprint: 701150e7b5508e76f9a10d5d22ef94f0
2026-01-08T18:01:55.318027Z [cowrie.ssh.transport.HoneyPotSSHTransport#debug] hex alg=b'curve25519-sha256' key alg=b'ssh-ed25519'
2026-01-08T18:01:55.318103Z [cowrie.ssh.transport.HoneyPotSSHTransport#debug] outgoing: b'aes128-ctr' b'hmac-sha2-256' b'none'
2026-01-08T18:01:55.318156Z [cowrie.ssh.transport.HoneyPotSSHTransport#debug] incoming: b'aes128-ctr' b'hmac-sha2-256' b'none'
2026-01-08T18:01:55.324061Z [cowrie.ssh.transport.HoneyPotSSHTransport#debug] NEW KEYS
2026-01-08T18:01:55.325066Z [cowrie.ssh.transport.HoneyPotSSHTransport#debug] starting service b'ssh-userauth'
2026-01-08T18:01:55.325752Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'root' trying auth b'none'
2026-01-08T18:01:55.325866Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'root' trying auth b'password'
2026-01-08T18:01:55.326033Z [HoneyPotSSHTransport,1,192.168.56.1] Could not read etc/userdb.txt, default database activated
2026-01-08T18:01:55.326152Z [HoneyPotSSHTransport,1,192.168.56.1] Login attempt [b'root'/b'ls'] succeeded
2026-01-08T18:01:55.326152Z [HoneyPotSSHTransport,1,192.168.56.1] Initialized emulated server as architecture: linux-x64-lsb
2026-01-08T18:01:55.326152Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'root' authenticated with b'password'
2026-01-08T18:01:55.326152Z [cowrie.ssh.transport.HoneyPotSSHTransport#debug] starting service b'ssh-connection'
2026-01-08T18:01:55.326152Z [cowrie.ssh.connection.CowrieSSHConnection#debug] got channel b'session' request
2026-01-08T18:01:55.326152Z [cowrie.ssh.session.HoneyPotSSHSession#info] channel open
2026-01-08T18:01:55.326152Z [cowrie.ssh.connection.CowrieSSHConnection#debug] got global b'no-more-sessions@openssh.com' request
2026-01-08T18:01:55.326152Z [twisted.conch.ssh.session#info] Handling pty request: b'xterm-256color' (30, 120, 640, 480)
2026-01-08T18:01:55.326152Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,1,192.168.56.1] Terminal Size: 120 30
2026-01-08T18:01:55.326152Z [twisted.conch.ssh.session#info] Getting shell
2026-01-08T18:02:02.521593Z [HoneyPotSSHTransport,1,192.168.56.1] CMD: ls
2026-01-08T18:02:02.522115Z [HoneyPotSSHTransport,1,192.168.56.1] Command found: ls
2026-01-08T18:05:02.110400Z [twisted.conch.ssh.session#info] exitCode: 1
2026-01-08T18:05:02.110578Z [cowrie.ssh.connection.CowrieSSHConnection#debug] sending request b'exit-status'
2026-01-08T18:05:02.110578Z [-] Closing TTY Log: var/lib/cowrie/tty/1d887ce8f8672e4914d9800e801cd74ecd805dd9366c0bc42dc16adc0197dc2f after 183.1 seconds
```

También lo podremos ver de una forma algo más ordenada si hacemos un nano cowrie.json



```
GNU nano 6.2 cowrie.json
{"eventid":"cowrie.session.connect","src_ip":"192.168.56.1","src_port":52891,"dst_ip":"192.168.56.10","dst_port":2223,"session":"b0295af73769","protocol":"ssh","message":"New
{"eventid":"cowrie.client.version","version":"SSH-2.0-OpenSSH_for_Windows_9.5","message":"Remote SSH version: SSH-2.0-OpenSSH_for_Windows_9.5","sensor":"srv-seguro","uid":"4
{"eventid":"cowrie.client.kex","hashh":"701158e75b508e76f0410d5d22ef9df0","hashhAlgorithms":"curve25519-sha256,curve25519-sha256@libssh.org,ecdh-sha2-nistp256,ecdh-sha2-nistp
{"eventid":"cowrie.login.success","username":"root","password":"hola","message":"login attempt [root/hola] succeeded","sensor":"srv-seguro","uid":"4529de08-ecb7-11f0-b6b7-61d9bdaa313c"}
{"eventid":"cowrie.client.size","width":120,"height":30,"message":"Terminal Size: 120 30","sensor":"srv-seguro","uid":"4529de08-ecb7-11f0-b6b7-61d9bdaa313c","timestamp":"2026-01-08T17:56:42.18
{"eventid":"cowrie.session.closed","duration":"194.2","message":"Connection lost after 194.2 seconds","sensor":"srv-seguro","uid":"4529de08-ecb7-11f0-b6b7-61d9bdaa313c","time
{"eventid":"cowrie.session.connect","src_ip":"192.168.56.1","src_port":54802,"dst_ip":"192.168.56.10","dst_port":2223,"session":"9a56ff49fb06","protocol":"ssh","message":"New
{"eventid":"cowrie.client.version","version":"SSH-2.0-OpenSSH_for_Windows_9.5","message":"Remote SSH version: SSH-2.0-OpenSSH_for_Windows_9.5","sensor":"srv-seguro","uid":"4
{"eventid":"cowrie.client.kex","hashh":"701158e75b508e76f0410d5d22ef9df0","hashhAlgorithms":"curve25519-sha256,curve25519-sha256@libssh.org,ecdh-sha2-nistp256,ecdh-sha2-nistp
{"eventid":"cowrie.login.success","username":"root","password":"1234","message":"login attempt [root/1234] succeeded","sensor":"srv-seguro","uid":"4529de08-ecb7-11f0-b6b7-61d9bdaa313c"}
{"eventid":"cowrie.client.size","width":120,"height":30,"message":"Terminal Size: 120 30","sensor":"srv-seguro","uid":"4529de08-ecb7-11f0-b6b7-61d9bdaa313c","timestamp":"2026-01-08T17:56:42.18
{"eventid":"cowrie.session.params","arch":"linux-x64-lsb","message":[],"sensor":"srv-seguro","uid":"4529de08-ecb7-11f0-b6b7-61d9bdaa313c","timestamp":"2026-01-08T18:01:59.01
{"eventid":"cowrie.command.input","input":"ls","message":"xcpd: ls","sensor":"srv-seguro","uid":"4529de08-ecb7-11f0-b6b7-61d9bdaa313c","timestamp":"2026-01-08T18:02:02.521593
{"eventid":"cowrie.session.closed","duration":"186.8","message":"Connection lost after 186.8 seconds","sensor":"srv-seguro","uid":"4529de08-ecb7-11f0-b6b7-61d9bdaa313c","time
{"eventid":"cowrie.session.connect","src_ip":"192.168.56.1","src_port":52707,"dst_ip":"192.168.56.10","dst_port":2223,"session":"16c419ebd2cd","protocol":"ssh","message":"New
{"eventid":"cowrie.client.version","version":"SSH-2.0-OpenSSH_for_Windows_9.5","message":"Remote SSH version: SSH-2.0-OpenSSH_for_Windows_9.5","sensor":"srv-seguro","uid":"4
{"eventid":"cowrie.client.kex","hashh":"701158e75b508e76f0410d5d22ef9df0","hashhAlgorithms":"curve25519-sha256,curve25519-sha256@libssh.org,ecdh-sha2-nistp256,ecdh-sha2-nistp
{"eventid":"cowrie.login.success","username":"root","password":"hola buenas","message":"login attempt [root/hola buenas] succeeded","sensor":"srv-seguro","uid":"4529de08-ecb7-11f0-b6b7-61d9bdaa313c"}
{"eventid":"cowrie.client.size","width":120,"height":30,"message":"Terminal Size: 120 30","sensor":"srv-seguro","uid":"4529de08-ecb7-11f0-b6b7-61d9bdaa313c","timestamp":"2026-01-08T18:02:02.521593
{"eventid":"cowrie.session.params","arch":"linux-x64-lsb","message":[],"sensor":"srv-seguro","uid":"4529de08-ecb7-11f0-b6b7-61d9bdaa313c","timestamp":"2026-01-08T18:12:31.77
{"eventid":"cowrie.session.closed","duration":"186.8","message":"Connection lost after 186.8 seconds","sensor":"srv-seguro","uid":"4529de08-ecb7-11f0-b6b7-61d9bdaa313c","time
```